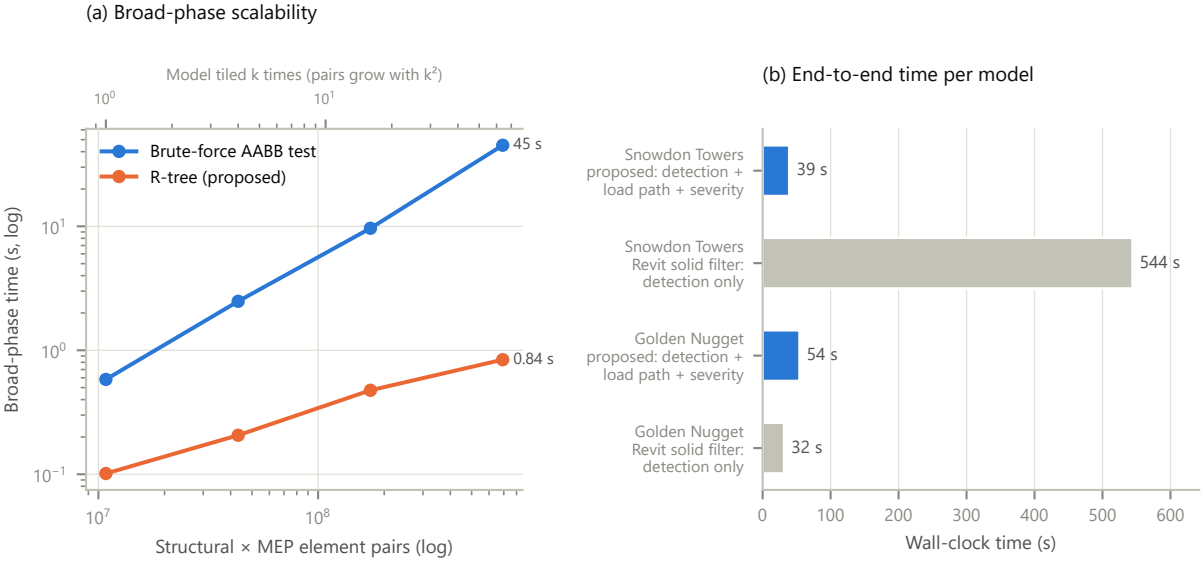
(a) Broad-phase scalability
Model tiled k times (pairs grow with k²)
10⁰
10¹
Brute-force AABB test
R-tree (proposed)
45 s
0.84 s
Broad-phase time (s, log)
10¹
10⁰
10⁻¹
10⁷
10⁸
Structural × MEP element pairs (log)
(b) End-to-end time per model
Snowdon Towers proposed: detection + load path + severity
39 s
Snowdon Towers Revit solid filter: detection only
544 s
Golden Nugget proposed: detection + load path + severity
54 s
Golden Nugget Revit solid filter: detection only
32 s
0
100
200
300
400
500
600
Wall-clock time (s)